1. **Healthcare and IoT Device Vulnerability Testbed**
2. **Team Members**

- Justin Bower - jbower2024@my.fit.edu
- Nathan Maloney - nmaloney2022@my.fit.edu
- Ipule Pipi - ipipi2022@my.fit.edu
- Jordan Chesley - jchesley2022@my.fit.edu

3. **Faculty advisor:** Dr. Snedah Sudhakaran and Dr. Abdullah Aydeger
4. **Client:** Faculty advisors and Prospective Manufacturers/Engineers
5. **Date(s) of Meeting(s) with the Client for developing this Plan:**
   8/24/26 @ 4:30 p.m. with Client in person, 8/26/26 @ 5:00 p.m. with Client in person, 8/27/26 @ 5:00 p.m. with Client on Zoom

6. **Goal and motivation:**

The goal of this project is to create a cybersecurity testbed for examining IoT and medical devices. This will allow us to easily probe for security vulnerabilities, for the benefit of researchers and manufacturers. By establishing a unified platform for reverse engineering and vulnerability research, researchers can more easily find exploits via a multitude of methods. The testbed will accelerate the discovery of weaknesses in device software, hardware interfaces, and communication protocols.

Currently, the IoT lab lacks such a unified platform for pulling binaries, examining firmware, and exploring other device exploits; moreover, we intend to satisfy this need by creating this testbed. The platform will support device connection, firmware extraction, entropy and decryption analysis, reverse engineering with industry-standard tools, automated and custom exploitation scripts, all through an intuitive client-side graphical interface. In addition, the tool will be integrated with a website for documentation and CVE tracking, with server-side support used where required, such as backend support and IoT specific needs.

This project will challenge our technical abilities, provide real world benefits, and further our knowledge of cybersecurity, integrating tools, automation, and other related skills. In doing so, it will equip teams that have limited experience in reverse engineering with a secure environment, furnished with automated analysis pipelines that directly support both academic and industry efforts. This will all be in service of the ultimate goal of strengthening the security of IoT and medical systems through vulnerability research.

7. **Approach (key features of the system):**

- Users can connect a variety of IoT and medical devices that use standard interfaces, wired or wireless, to the testbed, enabling full device access, interaction, and manipulation.
- Users can extract firmware directly from connected IoT and medical devices, obtaining complete firmware/binaries for detailed examination and vulnerability assessment.
- Users can analyze firmware entropy to evaluate randomness and compression characteristics
- Users can utilize industry-standard open-source tools to perform reverse-engineering tasks and comprehensive vulnerability research.
- Users can input their own CVE lists, and automatically retrieve lists of known CVEs, which are stored server-side in a database.
- Users can utilize pre-made scripts or upload custom exploitation scripts to rapidly test known , suspected, or new vulnerabilities, and potential attack vectors.
- Users can perform all connection, extraction, analysis, and exploitation tasks through an intuitive client-side graphical interface, thereby streamlining the process for both novice and experienced users.
- Users can create and upload documentation, or integrate existing documentation via GitHub, with all materials stored in server-side storage.

8. **Novel features/functionalities: Discuss which features/functionalities, if any, are novel and why.**

- A testbed that tightly integrates many disparate tools, automates firmware extraction, firmware analysis, entropy analysis, decryption, into one platform, is a novel approach to cybersecurity analysis.
- The testbed reduces the background knowledge needed for manufacturers or researchers to reverse-engineer Medical/IoT devices while unifying the reverse-engineering toolset.
- The testbed is specifically tailored for IoT and medical hardware, which is often proprietary and notoriously difficult to analyze.
- The testbed allows for direct firmware extraction from connected devices via standard interfaces and therefore streamlines the process of obtaining firmware/binaries for vulnerability assessment.
- The testbed allows for built-in entropy analysis combined with pre-existing decryption techniques to uncover encrypted or obfuscated firmware sections more efficiently.
- The testbed has support for both ready-made exploitation scripts and user-developed custom scripts provides flexibility to confirm known weaknesses or explore new attack vectors.

- The testbed will have backend/web support for supporting the local GUI that lowers the barrier to entry so that both novice and experienced users can perform complex tasks with greater ease.
- The testbed will have documentation storage and a CVE database with associated automations, accelerating the research process via targeted testing, and expedited sharing of results
- The testbed's benefits scale with the difficulty of reverse-engineering each device and contribute to a more secure future for IoT and Medical devices by making thorough examination more accessible.

9. **Algorithms and tools: Algorithms and software tools we are using/will be used**

   Overall Project/Multi-purpose
   Github: Version control and code collaboration tool
   VSCode: IDE for general development, testing, and demo purposes
   VSCode Dev Container: Standardized development environment
   Python: Ideal for scripting purposes and simple integrations with many tools via pre-existing libraries/APIs

   Communication:
   Email: Official correspondence and sending attachments
   Discord: More informal communication for day-to-day planning and coordination

   Reverse Engineering Tools:
   Ghidra: Open source tool for binary/firmware analysis
   Wireshark/Tshark (GUI and CLI): Open source tool for network analysis
   Binwalk CLI: Useful for examining files, extracting sub-files, and entropy analysis
   Foremost CLI: Useful for examining files, extracting sub-files, and entropy analysis
   GDB/pwndbg: Useful for low-level Assembly-language reverse engineering and vulnerability research
   Adafruit nRF51822-based Bluefruit LE Sniffer: USB device, drivers, and matching API for network capture (Bluetooth traffic logging as pcap files)
   Various Wireless Adapters: USB device, drivers, and matching API for network capture (Wifi/RF traffic logging as pcap files)
   EEPROM and Live Chipset Reader: Useful for pulling firmware directly off PCBS
   Soldering irons: For connecting and disconnecting electrical connections
   Solder wiring (lead or non lead): For establishing and holding electrical connections
   Copper wick: For wiping unused/excess solder
   Alligator clips: For holding PCBs and providing temporary connections where soldering is unnecessary
   Overhead lights: To provide clarity when soldering and doing smaller electrical tasks

Frontend and Backend Tools:
React: Frontend of website
TailwindCSS: Styling of website
JavaScript/TypeScript: Website base language/API calling functionality
Terraform: Deployment of services to cloud in a reproducible fashion that allows for easy transfer of the stack to a different AWS account
Amazon Web Services (AWS): Provides web server hosting, database backend, and other miscellaneous compute needs listed below:

- Docker: Isolates tasks for security, ease of deployment and platform management
- DynamoDB: Database architecture backbone; keeps data organized, minimizes storage space complexity and computational overhead via data structuring
- S3 Object Storage: Simple artifact storage and web hosting
- Instance Lambda Functions: Used to optimize speed of repetitive tasks and minimize memory space complexity
- Durable Lambda Functions: Allows for easier control of long running cloud analysis that may involve waiting for user input
- API Gateway: Allows for a secure connection between local and cloud applications
- Cloudfront: Improved web latency and increased max traffic capacity
- Cognito: AWS managed sign on portal with authentication tokens and account management

10. **Technical Challenges: Discuss three main CSE-related challenges (for example, "we plan to use/do javascript for web programming, but we don't know much about javascript").**

One major challenge is our limited experience decrypting and deobfuscating firmware extracted from IoT and medical devices, many of which employ proprietary encryption, compression techniques, or custom formats that require extensive research and experimentation. This is further complicated by the PCBs reversed in semester 1, which lacked dedicated UART/JTAG pins or connectors. Therefore, this will require live firmware extraction with procured EEPROM, direct-chip readers, and other related tools, to enable subsequent testing of custom and pre-existing scripts/ tooling.

A secondary challenge centers on creating secure tooling environments and whether cloud integrations will be required. Users expect sandboxed settings in which to run the attacks they develop, yet integrating these environments with cloud infrastructure makes isolation significantly more difficult, since we must achieve full security and functionality. The system must isolate scripts and processes effectively so that exploitation and analysis tools can be executed safely, without crashes or unintended effects on the host system, connected devices, or cloud infrastructure. Containers alone cannot do this, as we will be presenting this to researchers whose sole job is intentionally breaking/exploiting things. Our approach is the use of containers and a well-designed cloud architecture to manage them.

The third and overarching challenge is the complex integration, synchronization, and state management required to bridge a multitude of local reverse-engineering tools with the AWS cloud infrastructure. The testbed relies on coordinating custom and local tools with a diverse cloud backend. In addition, orchestrating the transfer of large firmware files, and complex vulnerability artifacts, requires asynchronous communication and end-to-end state tracking. Furthermore, we must engineer stringent state management protocols using attempt identifiers to handle edge cases (interrupted uploads, timeouts, etc.) and prevent stale or conflicting retries. This must all be done while ensuring strict authorization, so users can only access their securely isolated data, and analysis jobs.

11. **Milestone 4 (Sep 28th) itemized tasks [AWS Website and Starting Demos]:**

Justin - RE/VR, Cyber Tooling/Demos, Local GUI, Project Management
- Prepare a demo of the Adafruit device and pcap file extraction using Wireshark/Tshark
- Complete CVE Research for wifi chipsets and enter said CVEs into the database for demo
- Study I2C sensor and plan integration into project, such as an oximeter demo, and planning how it will interface in the main project
- Respond to professors emails, communicate/coordinate with team members, attend professor meetings, create, and write most of the Project Plan
- Wipe lab computer and install Linux (Dr. Sudhakaran gave permission already)

Nathan - Web/Database Backend and Demos
- Create the infrastructure to allow for cloud based analysis
- Prepare a demo of cloud based analysis of device attack results
- Integrate artifact storage and database with local and cloud infrastructure
- Create secure API endpoints to allow local and cloud applications to interact
- Setup router for testbed and interfacing devices in the network

Ipule - Web Front End and Cloud Backend Integration
- Complete the authenticated device management workflow for adding and viewing IoT/medical devices through the web dashboard.
- Integrate device creation and retrieval with AWS API Gateway, Lambda, DynamoDB, and Cognito so device records are associated with the authenticated user.

- Implement the end-to-end firmware upload workflow, including firmware reservation, SHA-256 metadata, presigned S3 upload, upload completion verification, and retry support for interrupted upload attempts
- Implement the firmware viewing interface so users can retrieve and display firmware metadata, upload status, filename, version, file size, and other associated information for each registered device.
- Strengthen firmware upload state management and authorization by using firmware/attempt identifiers to prevent stale or conflicting retries and ensure users can access only their own devices and firmware.

Jordan - RE/VR, Cyber Tooling/Demos, Local GUI

- Continue local GUI work so users can trigger the RE/analysis/exploitation tools
- Continue developing pre-made exploitation scripts for GDB/pwndbg (ret2*)
- Develop pre-made exploitation scripts for Ghidra and/or integrate existing Ghidra scripts
- Begin live firmware extraction using EEPROM/chipset readers
- Set up Linux machine with necessary tools

12. Milestone 5 (Oct 26) itemized tasks: [Partial test-Bed Setup in Lab and Intermediate Demo]

Ipule - Web Front End and Cloud Backend Integration

- Integrate the firmware-analysis workflow into the web dashboard so a user can select an uploaded firmware image and initiate an analysis job.
- Develop frontend interfaces for viewing analysis jobs, execution status, and generated results associated with a specific device and firmware version.
- Integrate the dashboard with the backend analysis APIs and storage architecture developed by the team, including retrieval of analysis outputs and artifacts.
- Improve device and firmware workflow state handling by displaying clear pending, ready, processing, completed, and failed states where applicable.
- Implement user interface protections and validation for analysis actions so jobs can only be started against valid firmware owned by the authenticated user.

Nathan - Web/Database Backend and Demos

- Create proof of concept docker/sidecar deployments are properly set up with the tooling of the DS18B20 I2C sensor work
- Starting working for DS18B20 I2C sensor (and similar sensor's) exploitation can be verified by a new user through their own replication of the pipeline
- Testing security of tooling and platform itself, as part of the containerization and isolation of environment

Justin - RE/VR, Cyber Tooling/Demos, Local GUI, Project Management

- Begin integrating Adafruit and Wireshark/Tshark
- Continue development of pre-made exploitation scripts for wifi/bluetooth exploitation
- Keep project/team members moving and ensure Professors (customers) are happy
- Maintain email/Discord communications constantly
- Create intermediate start-to-finish proof of concept demo

Jordan - RE/VR, Cyber Tooling/Demos, Local GUI

- Continue local GUI work so users can trigger the RE/analysis/exploitation tools
- Build out Ghidra-based reverse engineering workflows for extracted firmware, producing analysis outputs
- Begin integrate entropy analysis and file segmentation tools (Binwalk/Foremost)
- Continue development of pre-made exploitation scripts for wifi/bluetooth exploitation
- Begin I2C sensor integration into project

13. Milestone 6 (Nov 23) itemized tasks: [Final Test-Bed Setup in Lab and Full Demo]

Ipule - Web Front End and Cloud Backend Integration

- Complete the end-to-end web workflow from device registration and firmware upload through analysis execution and result retrieval.
- Integrate available reverse-engineering and vulnerability-analysis outputs into the dashboard, such as extracted firmware information, entropy results, discovered files, vulnerabilities, or generated artifacts provided by the analysis backend.
- Improve the dashboard user experience by organizing devices, firmware versions, analysis jobs, and outputs into a clear workflow suitable for researchers and manufacturers.
- Perform integration and edge-case testing of authentication, device ownership, firmware upload/retry behavior, analysis initiation, and result retrieval.
- Prepare the web application for the final project demonstration by resolving integration issues, improving error handling, and documenting/demoing the complete user workflow.

Jordan - RE/VR, Cyber Tooling/Demos, Local GUI

- Finalize device connections, to firmware extractions, to entropy/decryption, to exploitation/analysis paths, creating multiple local pipelines that feeds results to the web backend
- Finalize local GUI work so users can trigger the RE/analysis/exploitation tools locally
- Complete and test the exploitation scripts against real device/firmware data

- Prepare demos of the cyber tooling integrations for the final presentation
    - GDB/pwndbg
    - Ghidra demo

Justin - RE/VR, Cyber Tooling/Demos, Local GUI, Project Management

- Finalize CVE entries and RE documentation for the final demo
- Prepare demos of the cyber tooling integrations for the final presentation
    - Adafruit/Wireshark
    - Binwalk/foremost
- Prepare a full end-to-end RE demo (through aforementioned local pipelines) for the final presentation as a video recording
- Keep project/team members moving to ensure Professors are satisfied with progress
- Maintain email/Discord communications constantly with team membersF and Professors

Nathan

- Finalize AWS permissions and backend security using least-privilege access.
- Complete integration between local analysis tools, cloud services, S3, DynamoDB, and the web application.
- Finalize and test containerized cloud analysis and isolation.
- Perform end-to-end testing and prepare the cloud/backend workflow for the final demo.

14. **Task matrix for Milestone 4** (teams with more than one person)

Team Member	Main Responsibility Category	Sub Task 1	Sub Task 2	Sub Task 3	Sub Task 4	Sub Task 5
Justin	Reverse Engineering, Vulnerability Research, Cyber Tooling/ Demos, Local GUI, Project management	Prepare a demo of the Adafruit device and pcap file extraction using Wireshark/ Tshark	Complete CVE Research for wifi chipsets and enter said CVEs into the database for demo	Study I2C sensor and plan integration into project, such as an oximeter demo, and planning how it will interface in the main project	Respond to professors emails, communicate/ coordinate with team members, attend professor meetings, create, and write most of the Project Plan	Wipe the lab computer and install Linux (Got permission already)
Nathan	Web/Database Backend and Demos	Create the infrastructure to allow for cloud based analysis	Prepare a demo of cloud based analysis of device attack results	Integrate artifact storage and database with local and cloud infrastructure	Create secure API endpoints to allow local and cloud applications to interact	Setup router for testbed and interfacing devices in the network
Ipule	Web Front End and Cloud Backend Integration	Complete the authenticated device management workflow for adding and viewing IoT/medical devices through the web dashboard.	Integrate device creation and retrieval with AWS API Gateway, Lambda, DynamoDB, and Cognito so device records are associated with the authenticated user.	Implement the end-to-end firmware upload workflow, including firmware reservation, SHA-256 metadata, presigned S3 upload, upload completion verification, and retry support for interrupted upload	Implement the firmware viewing interface so users can retrieve and display firmware metadata, upload status, filename, version, file size, and other associated information for each registered device.	Strengthen firmware upload state manageme nt and authorizati on by using firmware/a ttempt identifiers to prevent stale or conflicting retries and ensure users can

				attempts		access only their own devices and firmware.
Jordan	RE/VR, Cyber Tooling/ Demos, Local GUI	Continue local GUI work so users can trigger the RE/analysis/exploitation tools	Continue developing pre-made exploitation scripts for GDB/pwndbg (ret2*)	Develop pre-made exploitation scripts for Ghidra and/or integrate existing Ghidra scripts	Begin live firmware extraction using EEPROM/chip set readers	Install necessary tools on Linux machine

15. Approval from Faculty Advisors

- "I have discussed with the team and approved this project plan. I will evaluate the progress and assign a grade for each of the three milestones."
- Signature: ________________________________ Date: 8/28/2026